

## Solution Requirements

|                      |                                   |
|----------------------|-----------------------------------|
| <b>Date</b>          | 6 July 2026                       |
| <b>Team ID</b>       | XXXXXX                            |
| <b>Project Name</b>  | Personalized Networking Assistant |
| <b>Maximum Marks</b> | 4 Marks                           |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category             | Requirement Description                                                      | Priority (High/Medium/Low) |
|------|----------------------------------|------------------------------------------------------------------------------|----------------------------|
| 1    | User Profile Input               | <i>Allow users to enter their name, skills, interests, and career goals.</i> | High                       |
| 2    | AI Networking Message Generation | <i>Generate a personalized professional networking message.</i>              | High                       |

|   |                      |                                                                                 |        |
|---|----------------------|---------------------------------------------------------------------------------|--------|
| 3 | Theme Extraction     | <i>Detect important themes using the DistilBERT model.</i>                      | High   |
| 4 | Fact Checking        | <i>Verify topics using the Wikipedia API.</i>                                   | Medium |
| 5 | Conversation History | <i>Store generated networking messages in history.json.</i>                     | Medium |
| 6 | Feedback Collection  | <i>Save user feedback in feedback.json.</i>                                     | Medium |
| 7 | Recommendations      | <i>Provide personalized networking suggestions based on the user's profile.</i> | High   |
| 8 | User Interface       | <i>[Display all results using the Streamlit application.</i>                    | High   |

## Step 2: Non-Functional Requirements (NFR)

|   |                            |                                                            |                                          |
|---|----------------------------|------------------------------------------------------------|------------------------------------------|
| 1 | Performance & Speed        | <i>Generate networking suggestions quickly.</i>            | <i>Response time less than 3 seconds</i> |
| 2 | Scalability                | <i>Support multiple users and future cloud deployment.</i> | <i>Easily scalable using FastAPI</i>     |
| 3 | Security & Data Privacy    | <i>Store only necessary user information securely.</i>     | <i>Secure API communication</i>          |
| 4 | Reliability & Availability | <i>Application should operate without crashes.</i>         | <i>99% uptime during testing</i>         |
| 5 | Usability & Accessibility  | <i>Simple and easy-to-use interface for all users.</i>     | <i>Beginner-friendly Streamlit UI</i>    |

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|      |              |                         |                                     |

|   |                 |                                                 |                                |
|---|-----------------|-------------------------------------------------|--------------------------------|
| 6 | Maintainability | Code should be modular, readable, and reusable. | Well-structured Python project |
|---|-----------------|-------------------------------------------------|--------------------------------|